[Software Engineering 10th edition]

Web sections

Planning activities

Activity charts are a schedule representation that highlights the dependencies between project activities or tasks (Figure 1). The activity chart shows which activities can be carried out in parallel and which must be executed in sequence because of a dependency on an earlier activity.

Activities are represented as rectangles; milestones are shown with rounded corners. The tasks (rectangles) are annotated with the expected duration of the task in working days e.g. T3 is planned to complete in 15 working days. The milestones (rounded rectangles) are annotated with the date by which they are expected to be reached.

Each milestone is annotated by the date at which that milestone is expected to be reached. Dates in this diagram show the start date of the activity and are written in British style, where the day precedes the month. Remember that the dates are WORKING days, which I assume are Monday to Friday, with weekend excluded. Typically, a task starts on a Monday morning and finishes on a Friday evening.

Therefore, the date of completion is not simply found by adding the number of days to the starting date. Weekends have to be taken into account for tasks lasting more than 5 working days.

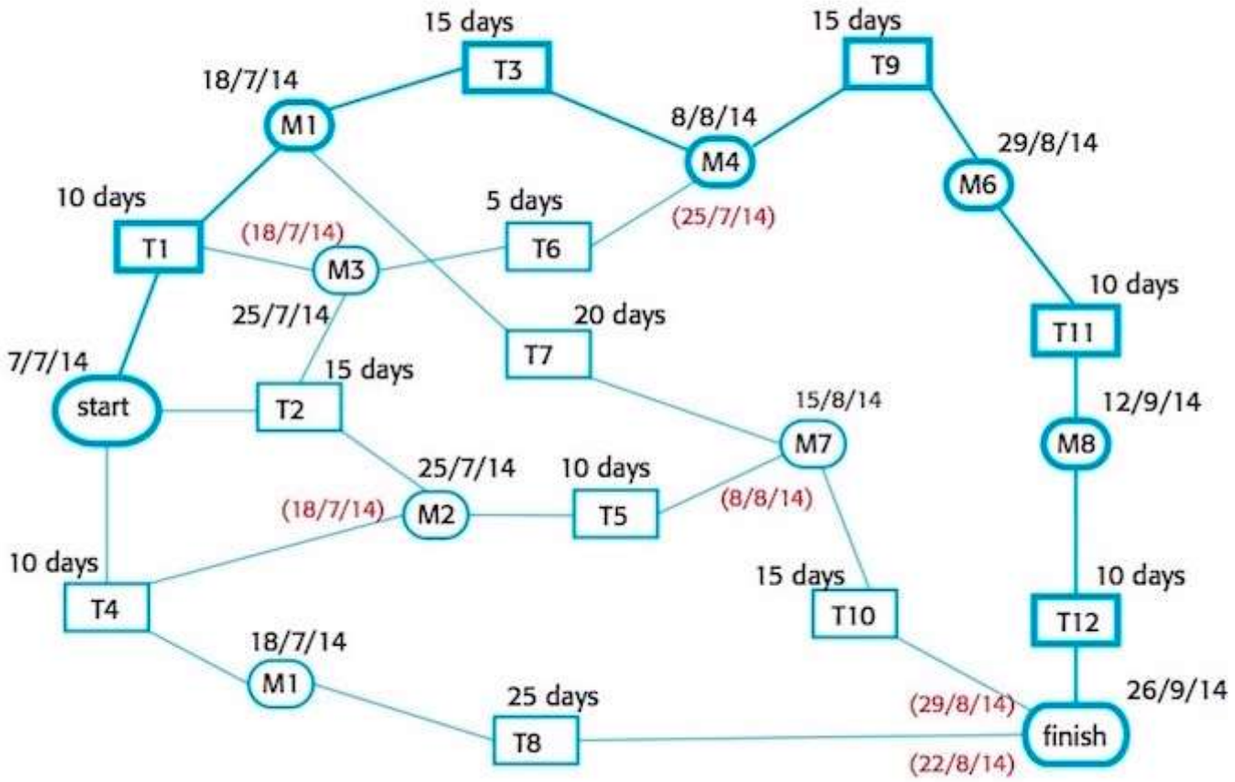


Figure 1: An example of an activity chart

Milestones may be annotated with more than one date. The date in black is the date at which the longest duration of tasks take e.g. milestone M4 is reached after T1 and T3 have completed on 8/8/14. However, milestone M4 also depends on the completion of T6, after T1 has finished. This is a shorter duration task (5 days) than T3, so the date shown in red is the planned date of completion of that task.

In practice, this means that task T6 could be delayed without affecting the overall schedule of the project, so long as it had completed by 8/8/14.

Before progress can be made from one milestone to another, all paths leading to it must be complete. For example, when activities T3 and T6 are finished and milestone M4 reached, then activity T9 can start.

Critical paths

The minimum time required to finish the project can be estimated by considering the longest path in the activity graph (the critical path). In this case, it is 12 weeks of elapsed time or 60 working days. The critical path is the sequence of dependent activities that defines the time required to complete the project.

In the activity chart for the project shown in Figure 1, the critical path is shown as a sequence of emboldened boxes (around the outside of the diagran, going clockwise).

The overall schedule of the project depends on the critical path. Any slippage in the completion in any critical activity causes project delays because the following activities cannot start until the delayed activity has been completed.

However, delays in activities that do not lie on the critical path do not necessarily cause an overall schedule slippage. So long as these delays do not extend these activities so much that the total time for that activity and future dependent activities does not exceed the critical path, the project schedule will not be affected. For example, if T8 is delayed by two weeks, it will not affect the final completion date of the project because it does not lie on the critical path. Most project management tools compute the allowed delays, as shown in the project bar chart.

Managers also use activity charts when allocating project work. They can provide insights into activity dependencies that are not intuitively obvious. It may be possible to modify the system design so that the critical path is shortened. The project schedule may be shortened because of the reduced amount of time spent waiting for activities to finish.

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